

MOHAMED LAAZIRI

Postdoctoral Researcher | Multiphysics Modeling | Scientific Computing | Nuclear Safety

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PROFESSIONAL PROFILE

Postdoctoral researcher in numerical modeling and scientific computing, specialized in numerical methods for multiphysics, thermo-hydro-mechanical and poromechanical problems in fractured media. Experienced in code coupling, finite element / finite volume / VEM / Nitsche methods, frictional contact, corium-concrete interaction and nuclear safety applications. Strong scientific publication record with hands-on experience in Fortran, C++, Python, MATLAB, CAST3M, ABAQUS, COMSOL and FreeFEM++.

TECHNICAL SKILLS

Modeling and simulation: multiphysics coupling, thermo-hydro-mechanical (THM) modeling, multiphase thermal hydraulics, corium-concrete interaction, fractured porous media, contact mechanics, concrete ablation, cracking.

Numerical methods: finite elements (FEM/P2), finite volumes, virtual element method (VEM), Nitsche formulations, mixed schemes, energy estimates, mesh adaptation, SUPG stabilization, Newton method.

Scientific programming: Fortran, C++, C, Python, MATLAB, R.

Software and codes: CAST3M, CIMAC, FreeFEM++, SIMULIA ABAQUS, COMSOL.

Systems and data tools: Linux, Kali Linux, Hadoop (HDFS/MapReduce), Spark, Yarn.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — Multiphysics Modeling for Nuclear Safety — CEA IRESNE, Cadarache, France | Dec. 2025 – Present

- Developing a multiphysics coupling between CIMAC (multiphase thermal hydraulics) and CAST3M (finite-element thermomechanics) to simulate corium-concrete interaction during severe nuclear accidents.
- Implementing numerical exchanges for heat fluxes, geometry evolution and source terms related to ablated concrete and degassing within the ANR IMMOG project.
- Contributing to improved prediction of concrete ablation, slab cracking and associated thermomechanical effects in reactor safety scenarios.

Doctoral Researcher — Applied Mathematics and THM Modeling — BRGM, ANDRA & Université Côte d'Azur, Laboratoire J.A. Dieudonné, Nice, France | Dec. 2022 – Nov. 2025

- Modeled thermo-hydro-mechanical processes in fractured porous media, coupling mechanics, flow and heat transfer.
- Developed and analyzed mixed, Nitsche and VEM discretizations for mixed-dimensional poromechanical models with frictional contact at matrix-fracture interfaces.
- Studied numerical schemes preserving energy estimates and thermodynamic consistency for coupled THM models.
- Published and presented research in Journal of Computational Physics, Computer Methods in Applied Mechanics and Engineering, ECMOR, FVCA and international conferences.

Master's Research Internship — Fractured Rock Mechanics and Flow — BRGM & Laboratoire J.A. Dieudonné, Nice, France | Mar. 2022 – Aug. 2022

- Adapted the Nitsche formulation to fractured/faulted rock mechanics coupled with flow in porous media.
- Compared the Nitsche approach with an existing mixed formulation implemented in a P2 finite-element prototype for mechanics.

Master's Research Internship — Numerical Simulation of Navier-Stokes Equations — Centre de Recherche en Automatique de Nancy (CRAN), France | May 2021 – Oct. 2021

- Simulated the 2D Navier-Stokes equations in MATLAB using the finite element method.
- Implemented a Newton method with SUPG stabilization and studied a two-level numerical method.
- Introduced observability and control concepts for the Navier-Stokes equations.

EDUCATION

PhD in Applied Mathematics — *BRGM, ANDRA & Université Côte d'Azur, Laboratoire J.A. Dieudonné, Nice, France* | Dec. 2022 – Nov. 2025

Dissertation topic: numerical modeling of thermo-hydro-mechanical processes in fractured porous media.

Master's Degree in Mathematics and Applications — Scientific Computing — *University of Reims Champagne-Ardenne, Reims, France* | Oct. 2021 – Sept. 2022

Finite elements, modeling, image processing, signal processing, machine learning and big data.

Master's Degree in Mathematics and Applications — Numerical Analysis of PDEs — *École Normale Supérieure, Casablanca, Morocco* | 2019 – 2021

Nonlinear PDEs, homogenization, asymptotic development, optimization, a posteriori estimates, mesh adaptation, flows in porous media and continuum mechanics.

Bachelor's Degree in Mathematics and Applications — Cryptography — *Hassan II University of Casablanca, Morocco* | 2016 – 2019

SELECTED PUBLICATIONS

- L. Beaudé, F. Chouly, **M. Laaziri** & R. Masson. Mixed and Nitsche's discretizations of Coulomb frictional contact-mechanics for mixed-dimensional poromechanical models. **Computer Methods in Applied Mechanics and Engineering**, **413**, 11612, 2023.
- L. Beaudé, F. Chouly, **M. Laaziri** & R. Masson. Mixed and Nitsche's discretizations of frictional contact-mechanics in fractured porous media. 14th International Conference on Large-Scale Scientific Computations, 2023.
- J. Droniou, **M. Laaziri** & R. Masson. Thermodynamically consistent discretisation of a Thermo-Hydro-Mechanical model. International Conference on Finite Volumes for Complex Applications, 265–273.
- J. Droniou, **M. Laaziri** & R. Masson. Discretisations of mixed-dimensional Thermo-Hydro-Mechanical models preserving energy estimates. **Journal of Computational Physics**, **515**, 113295, 2024.
- **M. Laaziri** & R. Masson. VEM-Nitsche fully discrete polytopal scheme for frictionless contact-mechanics. **SIAM Numerical Analysis**, 2025.
- M. Laaziri, R. Masson & A. Haidar. VEM-Nitsche fully discrete polytopal scheme for mixed-dimensional poromechanical models with Coulomb frictional contact at matrix-fracture interfaces. ECMOR 2024, 1–9.
- **M. Laaziri** & R. Masson. VEM fully discrete Nitsche's discretization of Coulomb frictional contact-mechanics for mixed-dimensional poromechanical models. **Computational Geosciences**, 2025.

CONFERENCES

SIAM Geosciences (Bergen, Norway, 2023); CFC23 (Cannes, France, 2023); FVCA (Strasbourg, France, 2023); Singular Days (Nice, France, 2022); MATHIAS Days – TotalEnergies (Paris, France, 2022); Workshop on Mathematical and Numerical Modeling of CO2 Storage (IFPEN, Paris, 2024); NEMESIS (Montpellier, France, 2024); ECMOR (Oslo, Norway, 2024); Mathematics and Decision Conference (Rabat, Morocco, 2024).

LANGUAGES

French | English | Arabic